Predictive Churn Analysis

Logistic Regression Classification & Customer Risk Scoring

Dataset: uploaded customer churn sample (15 records). The dataset contains 7 churned and 8 retained customers. Because the sample is very small, model metrics are intended as an internship demonstration and not as a production-quality model.

1. Data Preparation

CustomerID was excluded because it is an identifier. Churn was encoded as a binary target. Categorical features were One-Hot encoded; numeric features were median-imputed and MinMax-scaled. The data was split into 80% training and 20% testing using a reproducible random state and stratification.

2. Model

A Logistic Regression classifier with class balancing was selected because it is interpretable and appropriate for a binary churn-risk prediction workflow. The complete preprocessing and model steps are kept inside one pipeline.

3. Test-Set Evaluation

Metric	Result
Test records	3
Precision	1.000
Recall	1.000
F1-Score	1.000
ROC-AUC	1.000
Accuracy	1.000

4. Customer Churn Risk Scores

The final pipeline was refit on all 15 available customers and used to generate a churn probability for each customer. Risk levels are labelled Low (≤33%), Medium (34–66%) and High (>66%). The complete prediction file is included with the submission.

Customer	Actual Churn	Probability	Risk	Tenure	Contract
CUST-1003	Yes	87.4%	High	6	Month-to-Month
CUST-1014	Yes	86.7%	High	2	Month-to-Month
CUST-1007	Yes	86.5%	High	3	Month-to-Month
CUST-1010	Yes	85.2%	High	4	Month-to-Month
CUST-1005	Yes	83.2%	High	8	Month-to-Month
CUST-1012	Yes	79.8%	High	14	Month-to-Month
CUST-1001	Yes	79.4%	High	12	Month-to-Month
CUST-1006	No	33.6%	Medium	18	One Year

5. Business Interpretation

Customers receiving higher predicted churn probability can be prioritized for proactive retention actions such as onboarding support, contract-upgrade offers and faster resolution of repeated support issues. The accompanying risk-score CSV can be used as a simple retention-priority list.

6. Limitations

Only 15 observations are available. An 80/20 split leaves a very small test set, so precision, recall, F1 and ROC-AUC are unstable and should not be treated as production validation. A larger production dataset and cross-validation should be used before deploying a churn model.

Tools: Python, Pandas, Scikit-learn and ReportLab.